Logarithms and Their Properties

ClipSAT Printable Practice Suite (clipsat.org) • Topic 7.02

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

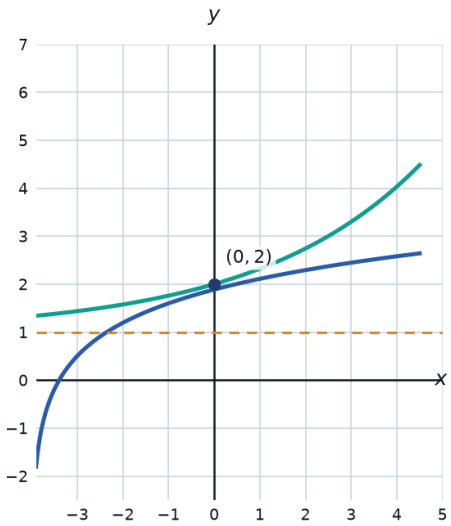
1 **[Foundational]** • **Target: Evaluating logarithms**

Evaluate each logarithm exactly:

a $\log_2 32$ **b** $\log_3 \frac{1}{9}$ **c** $\log 1000$ **d** $\ln e^7$

2 **[Foundational]** • **Target: Evaluating logarithms**

Use the change-of-base formula to evaluate $\log_5 18$, to three decimal places.



3 **[Foundational]** • **Target: Evaluating logarithms**

State the domain of $f(x) = \log(x - 4)$ and the inverse function of $g(x) = 2^x$.

4 **[Foundational]** • **Target: Properties of logarithms**

Expand fully using properties of logarithms: $\ln\left(\frac{x^3\sqrt{y}}{z^2}\right)$.

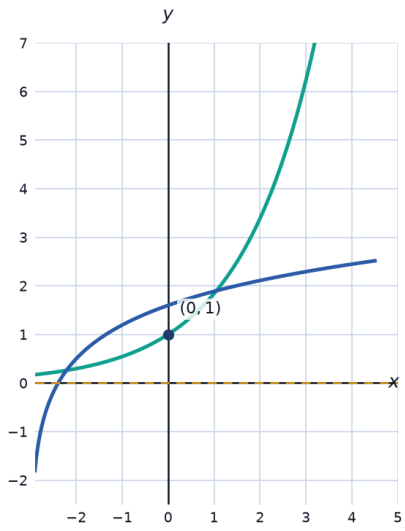
5 **[Foundational]** • **Target: Properties of logarithms**

Write as a single logarithm: $2\log x + 3\log y - \log z$.

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6 [Foundational] • Target: Properties of logarithms

Given $\log_b 2 = 0.431$ and $\log_b 3 = 0.683$, evaluate:



a $\log_b 6$ **b** $\log_b 8$ **c** $\log_b 1.5$

7 [Foundational] • Target: Graphs of logarithmic functions

For a curriculum assessment problem, suppose for $y = \log_2(x + 4) - 1$, state:

a the domain **b** the vertical asymptote **c** the x-intercept

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Graphs of logarithmic functions

Describe how the graph of $g(x) = -\ln(x - 2) + 3$ is obtained from $y = \ln x$.

9 [Intermediate] • Target: Logarithmic scales and applications

The Richter scale measures earthquake magnitude by $M = \log\left(\frac{I}{I_0}\right)$, where I is intensity and I_0 is a reference intensity.

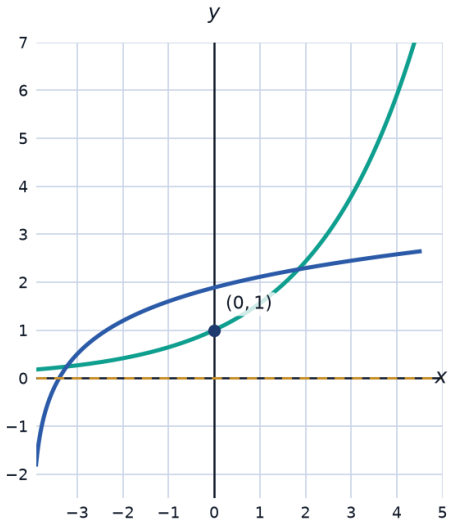
a An earthquake has $M = 6$. Express I in terms of I_0.

b How many times more intense is a magnitude-7 earthquake than a magnitude-5 earthquake?

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10 [Intermediate] • Target: Logarithmic scales and applications

Sound intensity level in decibels is $\beta = 10\log\left(\frac{I}{I_0}\right)$. If a sound's intensity doubles, by how much does β increase, to two decimal places?



11 [Intermediate] • Target: More practice with logarithm properties

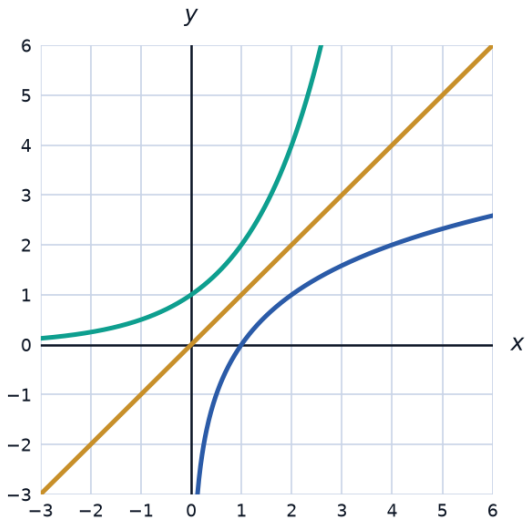
Expand fully: $\log_2\left(\frac{8x^2}{\sqrt[3]{y}}\right)$.

12 [Intermediate] • Target: More practice with logarithm properties

Write as a single logarithm and simplify: $\frac{1}{2}\ln 16 + \ln 3 - \ln 2$.

13 [Intermediate] • Target: Logarithms as inverse functions

The graphs of $y = 2^x$ and $y = \log_2 x$ are shown together with the line $y = x$. Explain why the two curves are reflections of each other across $y = x$.



14 [Intermediate] • Target: Solving simple logarithmic equations

Solve $2\log x = \log(x + 6) + \log 1$ for x (noting $\log 1 = 0$), and check for extraneous solutions.

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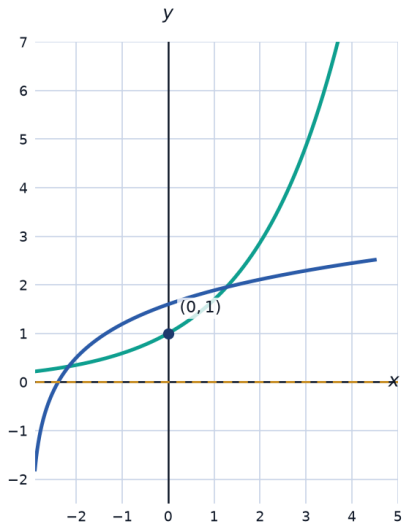
Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: The pH scale (a logarithmic application)** • **Cross-Lesson: 2.04 Law of Sines and Law of Cosines**

pH is defined as $pH = -\log[H^+]$, where $[H^+]$ is hydrogen ion concentration (mol/L). Find the pH of a solution with $[H^+] = 3.2 \times 10^{-5}$, to two decimal places.

16 **[Challenging]** • **Target: The pH scale (a logarithmic application)** • **Cross-Lesson: 3.01 The Unit Circle and Exact Values**

Two solutions have pH values of 3 and 6. Find the ratio of their hydrogen-ion concentrations $[H^+]$, and state which is more acidic.



17 **[Challenging]** • **Target: Common mistakes with logarithm properties** • **Cross-Lesson: 3.02 Fundamental Trigonometric Identities**

Explain why $\log(x + y) \neq \log x + \log y$ in general, using a specific numerical counterexample.

18 **[Challenging]** • **Target: Common mistakes with logarithm properties** • **Cross-Lesson: 3.03 Sum and Difference Formulas**

Simplify $\log_2(4x) - \log_2(2x)$ using the quotient property, and verify with a specific value of x.

19 **[Challenging]** • **Target: Evaluating logarithms** • **Cross-Lesson: 3.04 Double-Angle Formulas and Trigonometric Equations**

[Functions & Their Graphs · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

20 **[Challenging]** • **Target: Properties of logarithms** • **Cross-Lesson: 4.01 Circles and Parabolas**

[Unit 2 · Exponential and Logarithmic Functions · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

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Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Evaluating logarithms

Consider the functions $f(x) = \frac{1}{x-2}$ and $g(x) = 3x + k$. If $f(g(x)) = \frac{1}{2x-1}$, find the value of k.

a. 3

b. 2

c. 1

d. -1

22 [Foundational] • Target: Properties of logarithms

Let $f(x) = \log_2(ax + b)$ and $g(x) = 3^x - 1$. Given that $f(g(x)) = x$ and the point $(2, 1)$ lies on the graph of $y = g(x)$, find the value of $a + b$.

a. 4

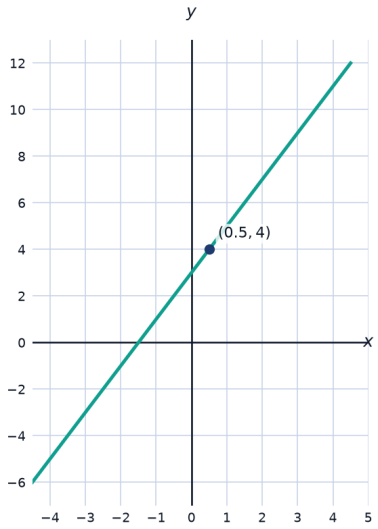
b. 3

c. 2

d. 1

23 [Foundational] • Target: Graphs of logarithmic functions

Given the function $f(x) = 2x + 3$. What is the value of $f^{-1}(7)$?



a. 1

b. 2

c. 3

d. 4

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24 [Foundational] • Target: Logarithmic scales and applications

Consider the functions $f(x) = 3x + 2$ and $g(x) = x - 5$. What is the value of $f(g(3))$?

a. 1

b. -4

c. 13

d. -10

25 [Foundational] • Target: More practice with logarithm properties

Which is the inverse of $f(x) = \log_2 x$?

a. $g(x) = \log_2 x$

b. $g(x) = 2x$

c. $g(x) = x^2$

d. $g(x) = 2^x$

26 [Foundational] • Target: Logarithms as inverse functions

If $f(x) = x^2 + 1$ and $g(x) = -x - 1$, find the sum of all solutions to $(f \circ g)(x) = 5$.

a. -2

b. 4

c. -6

d. -10

27 [Foundational] • Target: Solving simple logarithmic equations

Find the number of real solutions to $f(x) = x$ where $f(x) = x^3 - 3x$.

a. 1

b. 0

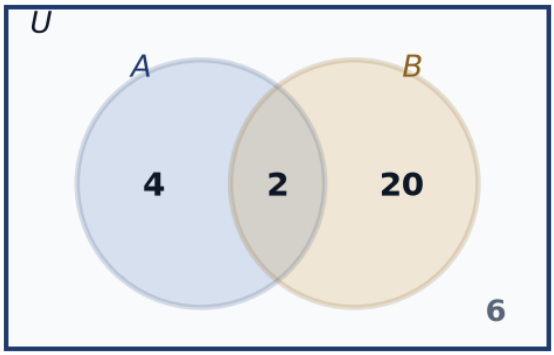
c. 3

d. 2

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28 [Foundational] • Target: The pH scale (a logarithmic application)

Find the number of surjections from a 4-element set to a 2-element set.



a. 14

b. 8

c. 16

d. 12

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Common mistakes with logarithm properties

Find all functions f with $f(x)^2 = x^2$ for all x.

a. $f(x) = x$, $f(x) = -x$, or $f(x) = |x|$ or $f(x) = -|x|$

b. $f(x) = x$ only

c. $f(x) = 0$

d. $f(x) = \pm 1$

30 [Intermediate] • Target: Evaluating logarithms

Find all polynomials $p(x)$ of degree 2 satisfying $p(x+1) - p(x) = 2x + 1$ and $p(0) = 0$.

a. $p(x) = 2x$

b. $p(x) = x^2 - x$

c. $p(x) = x^2 + x$

d. $p(x) = x^2$

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31 [Intermediate] • Target: Properties of logarithms

Solve: $3^x = 2^{x+1}$. Use logarithms.

a. $x = \frac{\ln 2}{\ln 3}$

b. $x = 1$

c. $x = \frac{\ln 3}{\ln 2}$

d. $x = \frac{\ln 2}{\ln 3 - \ln 2}$

32 [Intermediate] • Target: Graphs of logarithmic functions

What value must be excluded from the domain of $f(x) = \frac{1}{3x+3}$?

a. 3

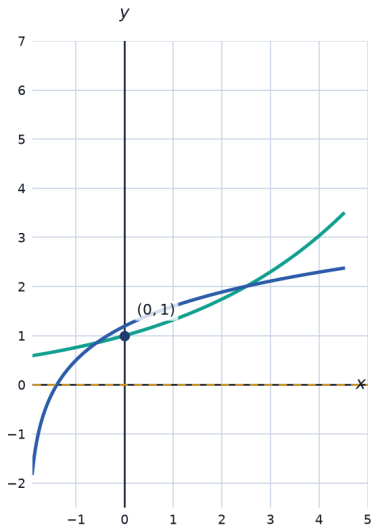
b. -1

c. -4

d. 2

33 [Intermediate] • Target: Logarithmic scales and applications

State and apply the intermediate value theorem: if $f(1) = -3$ and $f(4) = 5$, continuous on $[1, 4]$, then:



a. No conclusion possible

b. There exists $c \in (1, 4)$ with $f(c) = 0$

c. $f(2) = 1$

d. f is constant

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34 **[Intermediate]** • **Target: More practice with logarithm properties**

If $f(x) = -2x + 1$ and g is the inverse of f, find $g(f(-1))$.

a. 4

b. -1

c. 2

d. -10

35 **[Intermediate]** • **Target: Logarithms as inverse functions**

Find a value of c making f continuous at $x = 2$, where $f(x) = \{cx + 1, x < 2; x^2 - 1, x \geq 2$.

a. $c = 1$

b. $c = 3$

c. $c = 0$

d. $c = 2$

36 **[Intermediate]** • **Target: Solving simple logarithmic equations**

The Banach fixed-point theorem guarantees a unique fixed point when f is:

a. Any continuous function

b. Lipschitz with $L = 1$

c. A bijection

d. A contraction mapping on a complete metric space

37 **[Intermediate]** • **Target: The pH scale (a logarithmic application)**

Solve: $2^{x+1} = 5$. Express x using logarithms.

a. $x = \frac{\ln 5}{\ln 2} - 1$

b. $x = \frac{\ln 5 - 1}{\ln 2}$

c. $x = \log_2 5 - 1$

d. Both A and C

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Common mistakes with logarithm properties • Cross-Lesson: 3.03 Sum and Difference Formulas

Find all rational functions $f(x) = \frac{ax+b}{cx+d}$ that are their own inverse.

a. $b = c$

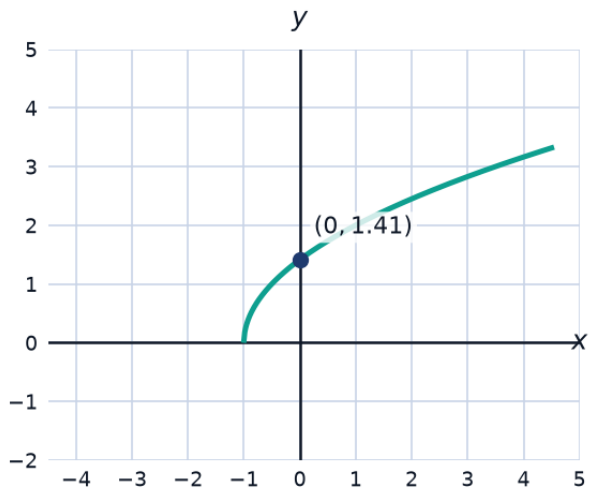
b. $f(f(x)) = x$ impossible for rational

c. $a = d$

d. $a = -d$ (i.e. $a + d = 0$)

39 [Challenging] • Target: Evaluating logarithms • Cross-Lesson: 3.04 Double-Angle Formulas and Trigonometric Equations

What is the smallest integer in the domain of $f(x) = \sqrt{2x+2}$?



a. -8

b. -9

c. -1

d. -10

40 [Challenging] • Target: Properties of logarithms • Cross-Lesson: 4.01 Circles and Parabolas

If $f(x) = \sqrt{9 - x^2}$, what is the shape of its graph?

a. Parabola

b. Upper semicircle of radius 3

c. Full circle

d. Hyperbola

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41 [Challenging] • Target: Graphs of logarithmic functions • Cross-Lesson: 4.02 Ellipses

Find the interval where $f(x) = x^2 - 4x + 3$ is positive.

a. $x > 0$

b. $1 < x < 3$

c. $x < 0$

d. $x3$

42 [Challenging] • Target: Logarithmic scales and applications • Cross-Lesson: 4.03 Hyperbolas

The Weierstrass function is continuous everywhere and:

a. Differentiable everywhere

b. Differentiable at integers only

c. Differentiable nowhere

d. Piecewise linear

43 [Challenging] • Target: More practice with logarithm properties • Cross-Lesson: 5.01 Vectors: Components and Operations

Which describes a vertical stretch of $y = f(x)$ by factor 4?

a. $y = f(4x)$

b. $y = f(x/4)$

c. $y = f(x) + 4$

d. $y = 4f(x)$

44 [Challenging] • Target: Logarithms as inverse functions • Cross-Lesson: 5.02 The Dot Product

Which function is the inverse of $f(x) = e^x$?

a. $g(x) = x^e$

b. $g(x) = \ln x$

c. $g(x) = 10^x$

d. $g(x) = \log x$

45 [Challenging] • Target: Solving simple logarithmic equations • Cross-Lesson: 5.03 Parametric Equations

A function has zeros at $x = -1, 0, 2$ and positive leading coefficient. Sketch shows which end behavior?

a. Left: $f \to +\infty$; Right: $f \to -\infty$

b. Left: $f \to -\infty$; Right: $f \to +\infty$

c. Both ends to $+\infty$

d. Both ends to $-\infty$

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